

Name(s): _____

DATA 101 Assignment 11: Word frequency

Work in teams of 2

Instructions. Create a new Jupyter Notebook. Use Markdown cells for all text explanations/headings and Code cells for Python code. Your file should be named `books.ipynb`. Once finished, export the notebook to HTML via **File → Save and export notebook as → HTML (.html)** and bring a printed copy to class.

1. Go to [Google Books Ngram Viewer](#). Choose two words and track their usage across the last two centuries. Why did you pick these two words? How did their usage frequency change? Why do you think they changed like that?
2. Go to Project Gutenberg's [Top 100 most downloaded books](#). Pick a book of your choice that we haven't analyzed in class. Clean stopwords and punctuations. In a table, report the following:
 - Top 10 most frequently used words and the number of times they occur
 - Total number of words
 - Total number of unique words
 - Type-token ratio
 - Average sentence length
3. Make a heatmap showing the co-occurrence counts of the Top 10 most frequently words you identified in the previous question.